ReviewPulse

Review Pulse v3.0 — Aspect-Based Sentiment Analysis

"One review, many opinions — sentiment for each aspect, with attention you can see."

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Problem · Plan & Risks

Torrens University Australia · DLE602 · Deep Learning · Assessment 2: Project Proposal · Dr. Tayab Din Memon

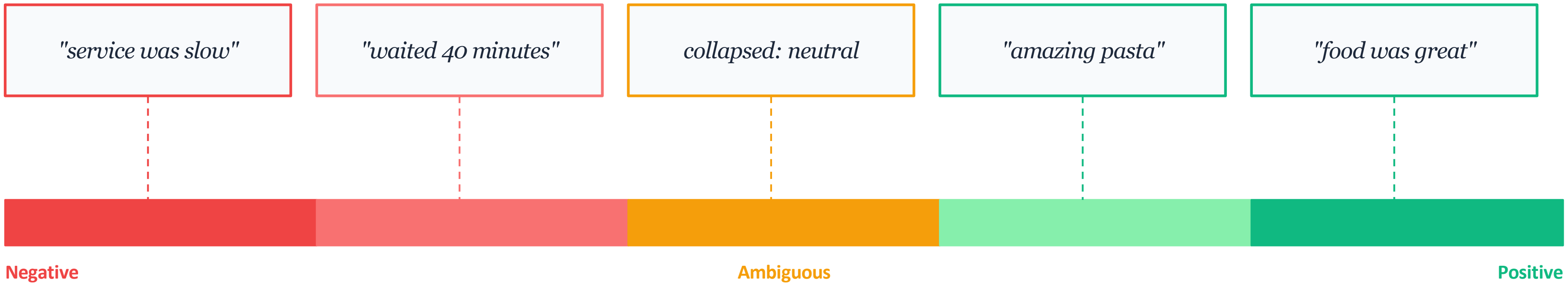
Why one label is not enough

"The food was great but the service was slow." One review, two opposite opinions — sentence-level models collapse it to a single neutral.

One review
two opposite opinions, one per aspect

Sentence-level models
our A1 N-gram and Zhao et al. CNN both collapse to one label

Business value
"food good, service bad" is actionable — "neutral" is not



Sentence-level collapse — opposite opinions average into a meaningless neutral

One limitation — a single label per text — solved step by step

1. **N-gram / lexicon era**
   local counts only · our A1 baseline
2. **CNN — Zhao et al. (2018)**
   deep but flat — one label per tweet
3. **ATAE-LSTM — Wang et al. (2016)**
   attention conditioned on the aspect
4. **BERT for ABSA — Sun et al. (2019)**
   contextual embeddings · auxiliary sentence trick
5. **LLM / neural discovery (2023–26)**
   zero/few-shot ABSA · heavy, opaque, costly
6. **The gap we target**
   light · explainable · aspect-level · faithful attention

THE GAP

Frontier is heavy

THEM

`LLM-driven · opaque`

↓

US

`light · faithful attention`

Why it matters

Few works pair a light, explainable model with faithful attention at the aspect level. That is exactly where we sit.

Four-model ladder — interpretability is the differentiator

LADDER 1–2 · BASELINES

Sentence-level floor

```
Ladder(
  1. TF-IDF + LogReg
  2. LSTM no target
  GloVe 100d
)
Note(
  no attention · no heatmap
)
```

ROLE

Set the sentence-level floor

LIMIT

One label per review — no why

LADDER 3 · ATAE-LSTM

Aspect-aware attention

Input (review, aspect)
aspect embedding + GloVe

LSTM + attention
aspect-aware attention weights

Attention weights
indicative token-level evidence

3-class output
positive · neutral · negative

ROLE

Explainable core — and the fallback if the transformer disappoints.

LADDER 4 · DISTILBERT

Fine-tuned transformer

Input (review, aspect)
aspect as auxiliary sentence (Sun et al., 2019)

Pretrained transformer
contextual representations · lower compute than BERT

Token attribution
or attention visualisation

SCOPE

1. Restaurants (core) + Laptops (extension) — pre-annotated
2. Topic modelling for aspect discovery — gated stretch goal

Only the aspect-conditioned models produce token-level evidence — indicative, not a causal explanation.

Modules 8–12: feasible and de-risked

Timeline: data + baseline → ATAE-LSTM → DistilBERT → interpretability + Streamlit demo → evaluation + report.

Small data → overfitting
RISK

Mitigated: dropout · early stopping · transfer learning from pretrained embeddings.

Heavy fine-tuning
RISK

Mitigated: DistilBERT fine-tuned on a free Colab GPU — zero-cost budget.

Scope creep
RISK

Mitigated: aspect extraction & topic modelling are gated stretch goals.

Pre-annotated data
WIN

SemEval-2014 ships with aspect labels — the feasibility win: no manual labelling.

Light compute
WIN

Free Colab · open datasets · open-source libraries — zero-cost end to end.

Working fallback
WIN

If the transformer underperforms, the ATAE-LSTM path still meets the rubric.

From one score per review to a transparent, per-aspect read

PROPOSED **Review Pulse v3.0 — built and demoed live in Assessment 3**

ASPECT-LEVEL

1

Sentiment per thing

Per-aspect labels instead of one collapsed score per review.

EXPLAINABLE

2

Attention heatmap

Shows which words drove each per-aspect call — the why, not just the label. Our differentiator.

COMPARED

3

Four-model ladder

TF-IDF → LSTM → ATAE-LSTM → DistilBERT, evaluated on identical benchmarks — accuracy and efficiency.

LIVE DEMO

4

Streamlit app

Working demo in Assessment 3 — type a review, watch per-aspect sentiment and attention live.

ReviewPulse

Questions?

2
SemEval datasets

2
models compared

5
modules · 8 to 12

1
live demo in A3

References: Wang et al. (2016) · Sun et al. (2019) · Devlin et al. (2019) · He et al. (2017) · Pontiki et al. (2014) · Hua et al. (2024) · Zhao et al. (2018) — full APA list in the written proposal.

GitHub **github.com/lfariabr/review-pulse** · ReviewPulse v3.0 · DLE602 Assessment 2